

The harp seal, *Pagophilus groenlandicus* (Erxleben, 1777). III. The underwater audiogram¹

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A free-field underwater audiogram from 0.76 to 100 kHz was obtained for *Pagophilus groenlandicus*. Areas of increased sensitivity occurred at 2 and 22.9 kHz. The lowest threshold was -32.9 db/ μ bar at 15.0 kHz. Above 64 kHz the threshold increases at a rate of 40 db/octave. The audiogram was similar to that of the *Phoca vitulina*. The effects of ambient noise on the audiogram are discussed.

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L'audiogramme obtenu sous l'eau, en champ libre (0.76-100 kHz) de *Pagophilus groenlandicus*, a révélé qu'il existe des zones de sensibilité accrue à 2 et 22.9 kHz; le seuil le plus bas est de -32.9 db/ μ bar à 15.0 kHz. Au-dessus de 64 kHz, le seuil s'élève à raison de 40 db/octave. L'audiogramme obtenu est semblable à celui de *Phoca vitulina*. On discute des conséquences, sur l'audiogramme, du bruit ambiant.

Introduction

The harp seal, *Pagophilus groenlandicus*, produces underwater phonations and high frequency clicks (Møhl and Ronald 1969). The latter may be useful for echolocation. Sound can carry information only at frequencies and intensities distinguishable to the listener. The basic attribute of hearing is sensitivity throughout the frequency range. As the harp seal is an amphibious mammal, the sensitivities of hearing in air (Terhune and Ronald 1971a) and underwater were studied.

Materials and Methods

A 4-year-old, immature, 90-kg, female harp seal that had previously been used to determine the air audiogram (Terhune and Ronald 1971a) was used to establish the free-field underwater audiogram. This was determined by a method of yes-no response to variable stimuli using methods described by Møhl (1968) and Terhune and Ronald (1971a), and a statistical analysis described by Wetherill and Levitt (1965).

A sine wave, originating from a Wavetek 112 signal generator, could be passed through a photoresistor gate to an attenuator (Fig. 1). The attenuator decreased the signal by set amounts, i.e., 0 decibels (db), or -12 to -24 db (in -2 db steps) or -60 db (no sound or catch trial). The resultant signal was then amplified and, to remove extraneous noise, was passed through a Dawe 1471 band pass filter. The signal was then broadcast by an Atlantic Research LC-32 hydrophone. The transmitting (Tx) hydrophone was located 0.5 m under the water surface and 0.6 m from the lower end of a plexiglass switch (No. 3). This switch was located centrally within a plastic resin-coated, wooden tank 3 × 5 × 1.5 m deep.

The seal was trained by operant techniques (Reynolds 1968) to voluntarily push switch No. 3. While this switch was depressed, the photoresistor gate opened and the preset signal, passed through the attenuator, was produced by the hydrophone. If the seal detected a signal, it was trained to push a response switch (No. 1) located some distance away. If the seal did not detect a signal (catch trial or signal error) it pushed another response switch (No. 2). The seal was rewarded with a piece of fish for each correct response. Each error resulted in the seal immediately receiving a jet of compressed air (0.7 kg/cm²) directed towards the face, while still at the response switch (No. 1 or 2). The seal received 3.5% of its body weight in fish per day. It was fed only during training or testing sessions.

The thresholds (intensities detected 50% of the time) were determined using up and down rules of Wetherill and Levitt (1965). After each correct response to a signal, the intensity of the next signal to be presented was reduced by one 2-db step. After each incorrect response to a signal, the intensity of the next signal to be presented was increased by one 2-db step. Thus, the sound intensity decreased until the seal could no longer detect it and then increased until the seal made the correct response. The highest intensities reached (hills) are just above the "momentary" threshold. The lowest intensities reached (valleys) are just below the "momentary" threshold. In an actual test session, the seal was first presented with above-threshold intensities. These were decreased until the seal made an error. After the first error had been made, the next six highest (hills) and six lowest (valleys) intensities were averaged to calculate the threshold (Wetherill and Levitt 1965). The standard deviation gives a measure of the range of the "momentary" thresholds about the mean threshold.

The seal was presented with an equal chance of being exposed to a signal or a catch trial. The order of the signals was based on Gellerman's schedule (1933). The maximum number of consecutive similar presentations was two. There were two testing sessions per day and they each required a total of 50-100 trials.

¹Migration Series 007.

After each test, the intensity and waveform of the signals were determined. A calibrated receiving (Rx) Atlantic Research LC-32 hydrophone was placed exactly where the seal's head had been while it pressed the switch No. 3. The received signal passed from the hydrophone to a

Tektronix FM 122 preamplifier, to a Krohn-Hite 3550 band pass filter and then to a Bruel and Kjaer 2409 voltmeter and a Tektronix 422 oscilloscope.

The underwater audiogram was determined from 0.76 to 100 kiloHertz (kHz) in half-octave steps. The upper

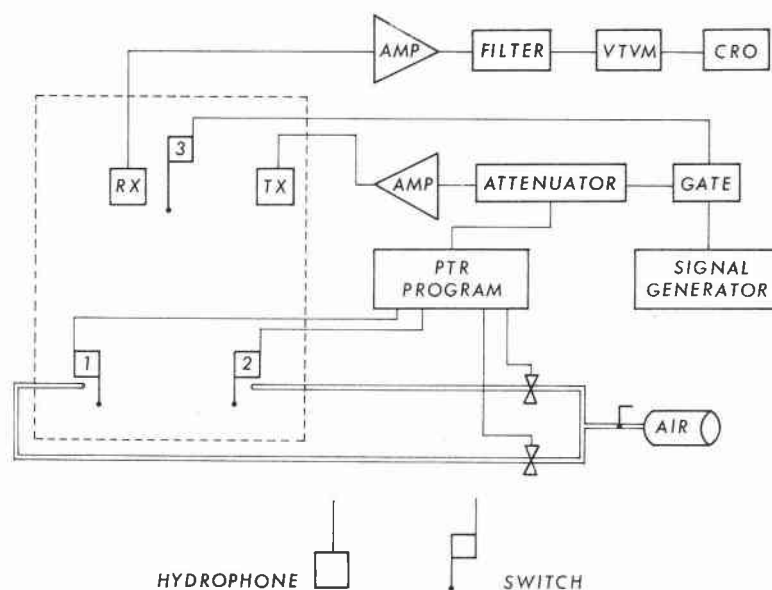


FIG. 1. Test equipment, underwater audiogram of a harp seal. For explanation see text.

TABLE 1
Acoustic sensitivity of a harp seal underwater

Frequency, kHz	Threshold, db/ μ bar	Standard deviation, db	Ambient noise level,* db/ μ bar	Catch trials, % correct	Date
0.76	-23	2.0	-60	71	1-3-71
1.0	-22	2.2	-64	68	12-12-70
1.4	-31	1.6	-67	81	12-31-70
2.0	-32	1.8	-70	80	12-18-70
2.0	-31	3.3	-70	89	1-19-71
2.8	-32	2.4	-73	83	12-30-70
4.0	-25	2.3	-75	88	12-18-70
5.6	-26	3.0	< -77	97	12-29-70
8.0	-31	2.0	< -78	83	12-19-70
11.3	-31	2.1		95	12-28-70
16.0	-29	2.6		92	12-27-70
22.9	-37	2.4		100	12-28-70
22.9	-30	2.1		93	1-18-71
32.0	-27	2.5		89	12-11-70
32.0	-25	2.1		82	1-17-71
44.9	-24	1.6		91	12-8-70
55.0	-19	2.5		78	1-15-71
64.0	+3	2.0		83	12-3-70
90.0	+14	2.2		79	12-5-70
90.0	+14	2.2		77	1-15-71
100.0	+56	2.8		77	12-5-70

*At the spectrum level.

and lower frequency limits were imposed by the apparatus and not by the seal. Four determinations were repeated once and two additional determinations were made.

Results

At all frequencies, standing waves and echoes caused a 10- to 20-db variation in the sound field. The calibrations of the sound field were repeatable (± 3 db based on similar voltages to the transmitting hydrophone).

Throughout the testing periods (Table 1), the seal's behavior appeared quite consistent. The seal, while pushing switch No. 3, would often move its head horizontally so that its left side would form a 45° angle between the longitudinal axis and the hydrophone. Occasionally the seal would push switch No. 3 twice before going to either of the response switches (No. 1 or 2). Although thresholds were obtained at 0.5 and 128 kHz, the intensities could not be calibrated. The calibration of the 100 kHz threshold is subject to some error resulting from slight distortion of the waveform produced by the receiving amplifier.

The testing range of the attenuator was 14 db. The observer's prediction of the seal's threshold was often incorrect and the responses of the seal

exceeded the range limits of the attenuator. When this happened, the threshold could not be determined during that session.

Discussion

As the seal was rewarded by a correct response it was assumed that the seal would locate the maximum intensity within the fluctuating sound field (Johnson 1967). Head movements by the seal while pushing switch No. 3 and the standard deviations of the threshold estimations support this assumption. Johnson (1967) in his study of the underwater audiogram of the bottlenose porpoise, *Tursiops truncatus*, found a variation of up to 14.6 db (at 10 kHz) and as low as 0.7 db (at 120 kHz). A previous underwater harp seal hearing determination (by a different method) at 11.3 kHz differed by 4 db from the threshold determined by the present study (Terhune and Ronald 1971b). Thus, the experimental variation was not great. Total variability from the true audiogram due to all factors was estimated at ± 10 db or less.

For the harp seal the correct catch trials varied from 68 to 100%. This variation was not reflected

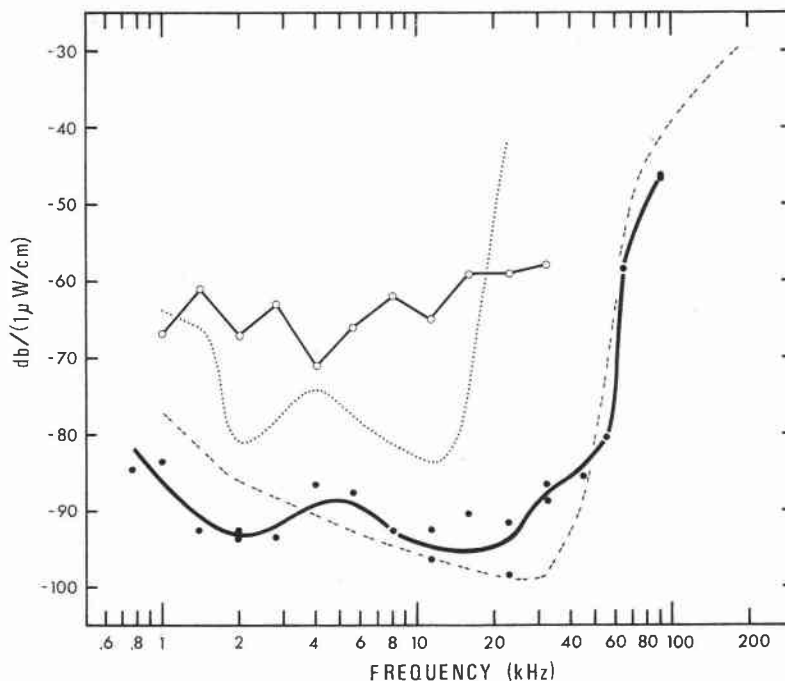


FIG. 2. Audiogram of a harp seal underwater. Harp seal underwater, ●; harp seal air, ○; harbor seal underwater (---); harbor seal air (...).

in the threshold values; the lower number of correct catch trials did not produce a threshold significantly lower than the preceding and following thresholds.

The underwater and air (Terhune and Ronald 1971a) audiograms of the harp seal (Fig. 2) were compared to those of the harbor seal, *Phoca vitulina* (Møhl 1968). The maximum sensitivity of the harp seal underwater was -32.9 db/ μ bar at 15.0 kHz. This intensity corresponds closely to 10^{-16} W/cm², a hearing sensitivity achieved by many mammals (Autrum 1948). The shape of the underwater audiogram of the harp seal is, in general, typical of that of a mammalian ear in its "adapted" medium. From the lower frequencies, the sensitivity shows an increase to its maximum and then decreases at the higher frequencies. Both the harp seal and the California sea lion,

Zalophus californianus (R. J. Schusterman 1971, personal communication) exhibit relative increases in sensitivity at the 2- and 12- to 24-kHz regions. There may actually be a decrease at 4 kHz. It is possible that this is a tuning or resonance phenomenon. The location of and (or) type of tuning cannot be determined without further studies.

The availability of a suitable subject and the time required often necessitate that determinations be taken only on a single animal (Johnson 1967; Møhl 1968; and Terhune and Ronald 1971a). The choice of an animal that has not had a record of illness or injury in the ear region reduces the likelihood of choosing an abnormal subject. There is close agreement between the two phocid audiograms (Fig. 2). This is not unexpected as the two species are very similar and

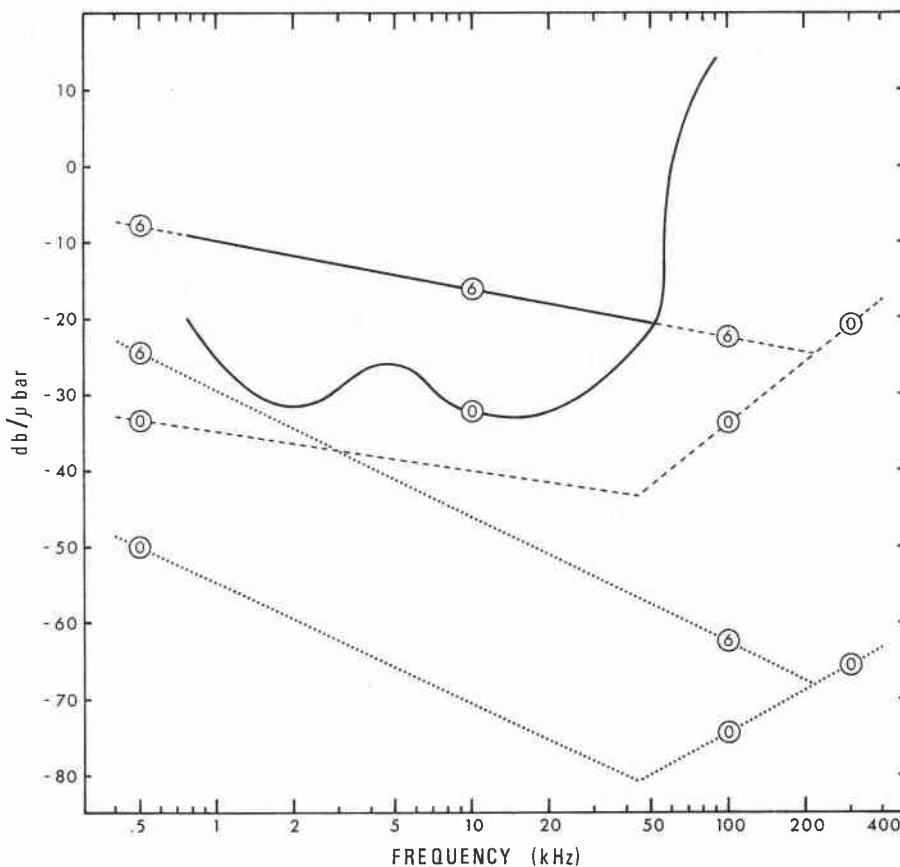


FIG. 3. Effects of ambient noise on the underwater hearing thresholds of a harp seal. Appropriate sea state (see text), O; harp seal underwater, threshold (—); ambient noise plus bandwidth effect (---); ambient noise (...).

often placed in the same genus (Doutt 1942). As the variations of the harp seal determinations were small and the maximum sensitivity was close to 10^{-16} W/cm², it is believed that the seal studied herein was a good representative of the species.

The air audiograms of both the harp and harbor seals (Fig. 2) show thresholds above their underwater counterparts. The sensitivities are expressed as power units (db/μW/cm²) rather than intensity units (db/μbar). This removes the effect of the differences in intensities caused by the different acoustic impedances of air and water. The decrease of sensitivity in air indicates that the seal ear is adapted to hearing underwater and is subject to some degree of acoustical impedance mismatch in air (Terhune and Ronald 1971a).

Ambient (background) noise levels will determine the limits of maximum sensitivity. Sound frequencies adjacent to a stimulus frequency mask the signal. The width of this interfering band is termed the critical bandwidth (Zwicker *et al.* 1957). The harp seal has a critical bandwidth of approximately 10% of the center or test frequency (Terhune and Ronald 1971a). It is the bandwidth of the listener that determines the actual effect of background noise on the threshold. In Fig. 3, the ambient noise levels for No. 0 and No. 6 sea states are shown at the spectrum level (Albers 1965). If ambient noise has similar properties to white noise (sound that contains all frequencies at a uniform intensity) then the effect of the harp seal's 10% critical bandwidth can be estimated. This approximation (Fig. 3) shows that at a No. 0 sea state, the seal's hearing threshold was unaffected by the ambient noise. The background noise of the test tank was below a No. 0 sea state, thus no part of the audiogram was masked. At a No. 6 sea state, the background noise definitely masks the threshold at the lower frequencies while frequencies above 50 kHz remain unaffected. From 0.76 to 50 kHz the masking by a No. 6 sea state is less than 20 db. Thus, the background noise, even during rough seas, is unlikely to hamper the underwater hearing abilities of harp seals.

The upper frequency discrimination of the harbor seal is 60 kHz (Mohl 1967) and it is likely that a similar value would hold true for the harp seal. The "click" phonation at 32 kHz (Møhl and Ronald 1969) approaches the most sensitive frequency. This would be a very apt frequency to use for echolocation or interanimal communication.

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